to decimate mosquito populations. One proven approach uses genetic modification to render the female mosquitoes flightless. The CRISPR/Cas9 genome editing method was used to introduce the genes to the Anopheles mosquitoes, and the genes are passed on to an astonishing 99 percent of the progeny. Once the female mosquitoes hatch, they cannot fly away and spread the diseases. The males can however, who go on to mate with other females, spreading the gene further. Tests in a lab environment have shown that a stable mosquito population can be eradicated in just a few months.

Researchers at the Johns Hopkins Bloomberg School of Public Health's Malaria Research Institute have used the CRISPR/Cas9 gene editing method to delete the FREP1 from the Anopheles mosquito. Knocking off the gene makes the mosquitoes resistant to the parasite Plasmodium. This is a new approach, which the research published only in March 2018. However, the team behind the experiment did realise that their were some concerns with the approach. The modified mosquitoes were not as fit as the unmodified ones, matured to adulthood more slowly, laid fewer eggs, and a smaller portion of the eggs were viable. However, the team is working on knocking off the gene only in the guts of the adult, so that the mosquito does not bear the fitness costs of editing the gene out. The experiment was successful at opening up new ways to tackle the problem, and study the parasites better.

Senior author of the paper, George Dimopoulos said in a press release, "Our study shows that we can use this new CRISPR/Cas9 gene-editing technology to render mosquitoes malaria-resistant by removing a so-called host factor gene. This gives us a good technological platform for developing advanced malaria-control strategies, based on genetically modified mosquitoes unable to transmit the disease, and for studying the biology of malaria parasites in their mosquito hosts."

Oxitec is a UK based company that uses gene editing as a form of pest control. They breed swarms of mosquitoes with a self-limiting gene. These mosquitoes are all male, and are released into the wild. They produce offspring with the females, which cannot mature to adulthood, and die off. Oxitec calls these mosquitoes "friendly mosquitoes". There are no toxins introduced to the ecosystem when this approach is used. The treatment is also specific to each species. Within the laboratory environment itself, an antidote is used to allow the friendly mosquitoes to reach adulthood.

Verily, an arm of Alphabet, uses a naturally occurring bacteria known as Wolbachia for its Debug Project. The mosquitoes are not genetically modified, but bred in a way that they are infected with the bacteria. The bacteria makes the male mosquitoes sterile.

A smart trap by Project Premonition.

Mating with the female mosquitoes does not result in the production of larvae. Large swarms of such mosquitoes can reduce the amount of breeding within the wild mosquito population. In July 2017, Verily released over 20 million such mosquitoes in California.

ROBOTS

Researchers need to identify the mosquitoes that have been introduced, and the mosquitoes in the wild. The regular way of doing this is to catch the mosquitoes in traps, and then process and study them. Conventional traps capture all insects indiscriminately, leaving them to be sorted by entomologists on the field – a labour intensive process. Project Premonition uses a whole bunch of high tech wizardry to accomplish the same thing.

The device is called a robotic smart trap. Each trap has 64 cells, embedded with sensors. The frequency of the insect wings beating is used to identify the insect. If the insect is a mosquito, it is trapped. When trapping the mosquito in this manner, the embedded sensors can also record the temperature, the lighting conditions and the time of the day. All of this helps understand the behavior of the mosquitoes more. Over nine species of mosquitoes have been trapped using this method. Additionally, the traps were also used to test the effectiveness of chemical lures. The traps had a success rate of 85 percent accuracy, when trapping mosquitoes. The data collected from the traps are processed by machine learning algorithms, to make better predictions of when these mosquitoes are likely to be active.

One of the roadblocks to introducing genetically modified mosquitoes into the wild, is to sort the males from the females. Almost all of the approaches introduce the genes in the wild through males. Robots help with this sorting process too.

Verily is developing automation technologies to segregate the males from the females. The technologies depend on image recognition. MosquitoMate, one of the collaborators of Verily, first passes the mosquito through a sieve, where the smaller males are isolated from the females. After this step, human intervention is again needed to "eyeball" and isolate any remaining females. This sorting needs to be automated considering the scale of modified mosquito deployment.

Helping humans in the fight against mosquitoes are robots, AI, smart traps, cloud computing and gene editing. Most of these campaigns are still in the experimental phase, but at least now, there is a sustained effort, by academics, startups, and large corporations, to tackle the growing menace of mosquitoes with the use of technologies.